AB/May-2026

PAPER CODE: BCE-C621

EVEN SEMESTER EXAMINATION (MAY-2026)

CLASS:B.TECH. SEMESTER:VI

PAPER TITLE: DISTRIBUTED SYSTEMS

Time: 3 hours Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks.

(5 x 6 = 30 Marks)

		CO	BL
Q-1	Discuss the role, functions, and importance of the Operating System in Distributed Computing systems	CO2	BL2
Q-2	Define events and notifications in distributed systems and discuss their roles with appropriate real-world examples.	CO2	BL4
Q-3	Explain the role of APIs in Inter-process communication.	CO1	BL1
Q-4	Explain client-server architecture with examples and diagrams.	CO3	BL3
Q-5	Explain the terms: 1) Replication. 2) Consistency. 3) Concurrency.	CO3	BL3
Q-6	Explain how the HTTP protocol works in web communication	CO1	BL1
Q-7	Explain about: 1) Race Condition. 2) Context Switching. 3) SOA.	CO4	BL2
Q-8	What is the purpose of logical clocks in distributed systems? How do they help in event ordering?	CO1	BL1
Q-9	How does CORBA facilitate communication among distributed objects?	CO4	BL4
Q-10	Briefly explain election algorithms and multicast communication	CO2	BL2

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any ***four*** questions in detail. Each question carries 10 marks.

(4 x 10 = 40 Marks)

		CO	BL
Q-1	Explain the concept of marshalling in distributed systems and discuss the challenges faced in different (heterogeneous) environments.	CO3	BL4
Q-2	Describe different system models in distributed systems and compare peer-to-peer and three-tier architectures with suitable examples.	CO2	BL2

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Q-3	Explain how DNS works as a name service and how it helps in fault tolerance and load balancing.	CO3	BL2
Q-4	Compare stateful and stateless file services in distributed systems with respect to performance, reliability, and scalability.	CO2	BL3
Q-5	Explain the role of events and process states in distributed systems and how they help in coordination and maintaining consistency.	CO1	BL1
Q-6	Discuss how caching, middleware, and load balancing improve the performance of web-based distributed systems.	CO4	BL2
Q-7	Compare Ricart-Agarwala and Maekawa algorithms for distributed mutual exclusion with their advantages and limitations.	CO1	BL1
Q-8	Explain different clock synchronization techniques such as Cristian's, Berkeley, and NTP, and compare their working.	CO2	BL3

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Paper Code